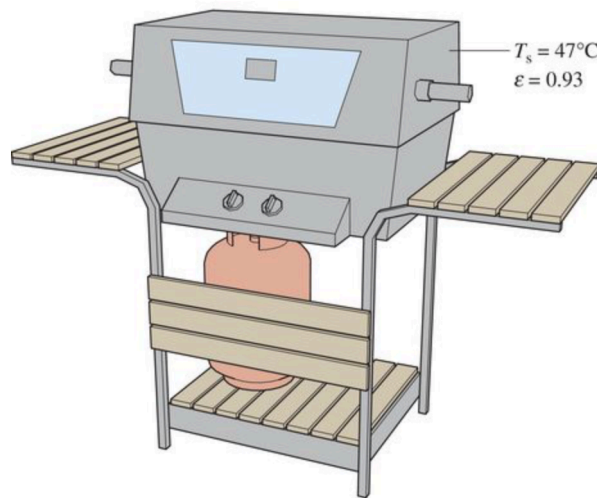


**CHE 201 HW 3**  
**Due: 11.59pm on 02/09/2020**

**1. (20 points)**

**2.41**  The outer surface of the grill hood shown in Fig. P2.41 is at  $47^\circ\text{C}$  and the emissivity is 0.93. The heat transfer coefficient for convection between the hood and the surroundings at  $27^\circ\text{C}$  is  $10 \text{ W/m}^2 \cdot \text{K}$ . Determine the net rate of heat transfer between the grill hood and the surroundings by convection and radiation, in kW per  $\text{m}^2$  of surface area.

$$T_0 = 27^\circ\text{C}$$
$$h = 10 \text{ W/m}^2 \cdot \text{K}$$



**FIGURE P 2.41**

**2. (20 points)**

Problem 2.45

**3. (20 points)**

Problem 2.49

**4. (20 points)**

Problem 2.54

**5. (20 points)**

Problem 2.57